

Detailed Project Report (DPR)

Agri-Biomass Biochar & Enriched Bio-Fertilizer — Modular Green-Category Micro-Units

Promoter: Arun T. (sole proprietary concern — no partnership) **Location:** Eruthenpathy, Chittur Taluk, Palakkad District, Kerala (Kerala-Tamil Nadu border) **Feedstock origin:** Own coconut-farm residues + open-market agri/wood biomass procured from multiple local sources (no single-supplier or partnership dependency) **Structure:** A rollout of **independent micro-units, each with fixed capital ≤ ₹50 lakh**, each designed to sit in the **Green** pollution category and use the **self-certification** consent route. **Prepared for:** Submission to District Industries Centre (DIC), Palakkad and lending bank (AIF / term loan) **Document status:** Planning-grade DPR — figures are indicative estimates to be firmed up with (a) a feedstock proximate-analysis lab report, (b) formal machinery quotations, and (c) a written KSPCB categorisation confirmation, before sanction.

1. Executive summary

This project establishes an **agri-biomass biochar and enriched (microbe-inoculated) organic bio-fertilizer** business as a **sole proprietary concern of Arun T.**, built as a series of **standalone micro-units, each ≤ ₹50 lakh in fixed capital**, on the promoter's own land at Eruthenpathy, Chittur, Palakkad.

- **Constitution:** sole proprietorship — **Arun T., proprietor**. There is no partnership and no partner-supplied feedstock.
- **Feedstock: multi-source, open-market and own-farm agri/wood biomass** — coconut fronds, petioles, husk, shells and spathe from the promoter's own farm; coir pith and shells from local dehuskers/coir units; and open-market residues (sawdust and wood chips, cashew shell, areca husk, paddy husk, groundnut shell, prunings). Not limited to tender-coconut husk and not dependent on any one supplier.
- **Pollution category — Green (target).** Each unit is designed as *general biomass biochar (soil-amendment) production*, which the KSPCB revised categorisation list (as on 29.12.2025) places in the **Green** category — distinct from "charcoal from coconut shell by pyrolysis" (Orange). See Section 7.
- **Unit size for lenient norms:** keeping each unit's **fixed capital ≤ ₹50 lakh** brings it within KSPCB's **self-certification** thresholds for Consent to Establish/Operate (circular dated 16.09.2025) and within the **micro-enterprise MSME** band.
- **Per-unit capacity:** **~0.5 t/day biochar** (~150 t/yr) from ~2 t/day mixed dry biomass; **~330 t CO₂/yr** removal.
- **Rollout: Unit-1 now**, replicated to **3 units** from cash flows / fresh micro-loans → aggregate **~1.5 t/day (~450 t/yr, ~990 t CO₂/yr)**.
- **Per-unit project cost:** **~₹52 lakh** (₹48 L fixed capital + ₹4 L working-capital margin); land is the promoter's own (in-kind, excluded from the ₹50 L capital test).
- **Financing (per unit):** promoter equity + **AIF term loan (3% interest subvention, 7 yrs)** + **CGTMSE** collateral-free cover + **Kerala ESS capital subsidy**.
- **Revenue stack:** enriched biochar sales (domestic + export) + **biochar carbon-removal credits** via an aggregator (Puro.earth CORCs).
- **Indicative payback:** ~2.5–3.5 years per unit (physical-sales base case); under ~2 years if carbon-credit revenue is realised.

2. Promoter

Promoter / constitution: **Arun T. — sole proprietary concern** (proprietorship). This is **not a partnership**; there is no "and Friends" or associate partner, and no partner-supplied raw material.

- Owns the project land — a 38-cent plot and an adjacent coconut farm (~200 m away) in Eruthenpathy, Chittur Taluk, Palakkad — which accommodates the phased micro-units and their feedstock yard.
- Contact: 4/296, Mangapallam South, RVP Pudur Post, Palakkad District, Kerala – 678555 · 9901955667 · er.arunt@gmail.com.
- Captive demand: the promoter's own coconut farm consumes and demonstrates the enriched product, de-risking market entry.

3. Project concept & rationale

Agri-biomass residues (coconut fronds/husk/shell, coir pith, wood/sawdust and other crop residues) are abundant, low-value and often burnt in the open. Converting them to biochar via controlled, low-emission pyrolysis:

1. Diverts local biomass waste from open burning into a productive use.

2. Produces a stable, carbon-rich **soil amendment**; enriched with organics + microbes it becomes a branded slow-release **bio-fertilizer** (the product is a soil input — **not** fuel charcoal and **not** activated carbon).
3. Generates durable **carbon-removal credits** (1 t biochar $\approx$ $\sim$ 2.2 t CO₂ sequestered for centuries).
4. Qualifies for strong central + state subsidy support as an agro-based / bio-fertilizer **micro** MSME.
5. Is structured to be **Green-category and self-certifiable**, so each unit can be established quickly and cheaply (Section 7).

4. Market analysis

Product (physical biochar), indicative prices:

Grade	₹/kg	25 kg bag
Raw bulk (agri)	15–40	₹375–1,000
Enriched / inoculated	50–120	₹1,250–3,000
Premium certified (EBC/organic)	up to 150	up to ₹3,750

Carbon credits: biochar CORCs trade ~US\$125–145 (Nasdaq CORCCHAR index); Indian industrial biochar clears ~€105–150/t CO₂. Biochar delivered ~43% of verified durable CDR volume; >90% of near-term supply is already contracted by buyers (Microsoft, Google, JPMorgan, Swiss Re) — a **supply-constrained, seller-favourable market**.

Demand strategy: anchor demand on (a) own-farm consumption, (b) enriched/branded regional sales, (c) export of the carbon credit via an aggregator. Sell *enriched* product (margin) rather than raw char (commodity).

5. Feedstock plan (multi-source, no partnership)

The unit is **not tied to a single feedstock or supplier**. It runs on a mix of own-farm and open-market biomass, giving supply security and reinforcing the *general biomass* (Green) classification.

Source	Material	Basis
Own coconut farm (200 m)	Fronds, petioles, husk, shells, spathe	Free / captive
Local dehuskers & coir units	Coir pith, coconut shells	Open-market purchase
Local sawmills / carpentry	Sawdust, wood chips, offcuts	Open-market purchase
Local farms / traders	Cashew shell, areca husk, paddy husk, groundnut shell, prunings, seasonal crop residue	Open-market purchase / collection

Two-site handling: the **farm (200 m)** is used for feedstock aggregation, a **sun-drying yard**, chipping/shredding and bulk storage (no processing); the **unit** carries only a 1–2 day day-bin, keeping bulk fuel away from the reactor (fire-safety and consent advantage).

Supply security: execute **supply MoUs / purchase arrangements with 3–4 local sources** (not one partner) plus own-farm residue — sufficient for both bank appraisal and carbon-credit additionality/traceability. Feedstock is documented **residual/waste biomass** for carbon additionality.

6. Technical design & capacity (per micro-unit)

6.1 Mass balance (design basis per unit: ~2 t/day mixed dry biomass)

Parameter	Value
Biomass feed (dried)	~2,000 kg/day
Biochar yield (~25–30% dry mass)	~0.5 t/day
Operating basis	~100 kg/hr feed → ~25 kg/hr biochar over ~20 h
Annual biochar (300 days)	~150 t/yr per unit
CO ₂ removal (~2.2×)	~330 t CO₂/yr per unit

Aggregate at 3 units: ~1.5 t/day biochar (~450 t/yr, ~990 t CO₂/yr).

6.2 Right-sized micro-unit ("small and replicable, not one big plant")

Unlike a single large plant, each unit is **right-sized to ~0.5 t/day** with its **own compact emission-control train** — no over-provisioning. Capacity is grown by **replicating whole units**, which keeps every unit independently $\leq ₹50$ L, Green and self-certifiable, and lets capital be staged.

Component (per unit)	Scale
Pyrolysis reactor / carbonizer	~0.5 t/day, continuous/semi-continuous, multi-feedstock
Emission control: afterburner/flare → cyclone → scrubber + mist eliminator → stack	~0.5 t/day, Green-compliant
Waste-heat feed dryer	~0.5 t/day
Char cooling, sieving, enrichment mixer, bagging	~0.5 t/day
Electrical, water, weighing, fire	micro-unit

6.3 Emission control & Green-category design

Closed, **continuous/semi-continuous** pyrolysis with a **syngas afterburner/flare** (self-fuelling the process and the dryer) → **cyclone/dust collector** → **alkaline scrubber with mist eliminator** → **stack** meeting KSPCB norms; **real-time temperature monitoring** and emergency shutdown; **Zero-Liquid-Discharge** (tar condensate and scrubber blowdown captured and reused/soaked into feed — no effluent to land or water). **No open burning** at any stage. The design is built to the **KSPCB "Environmental Guidelines for Charcoal Manufacturing Units" dated 02.12.2025** (emission control, stack height, ash/dust housekeeping, heat reutilisation, responsible feedstock sourcing).

7. Pollution-control category & consent strategy

Objective: establish each unit in the Green category and via self-certification — the lenient regime — legitimately, on the merits of the process, and by keeping each unit small.

7.1 Why Green applies

The KSPCB revised categorisation list **as on 29.12.2025** classifies *general charcoal / char (biochar) production* under **Green**, and distinguishes it from "*Charcoal Production from Coconut Shell by Pyrolysis*" — the **Orange** entry created by Notification No. KSPCB/159/2022-SEE-3 dated 10.08.2024. Our unit fits the **general / Green** description because:

- **Feedstock is mixed agri-biomass**, not a coconut-shell-only charge.
- **Product is biochar for soil**, not fuel charcoal and not activated carbon.
- **No pyrolysis-oil or carbon-black recovery** and **no chemical/steam activation** — the features that drive the heavier (Orange/Red) categories.
- **Low pollution potential** on the CPCB Pollution-Index basis — a closed process with afterburner, scrubber, controlled stack emissions and **Zero-Liquid-Discharge**.
- **Guideline-compliant** with the 02.12.2025 charcoal-manufacturing guidelines.

7.2 Why the $\leq ₹50$ lakh unit size matters

Category is set by the *activity*; the *unit size* governs the **consent process**. Keeping each unit's **fixed capital $\leq ₹50$ lakh** brings it within KSPCB's **self-certification** thresholds for CTE/CTO (circular dated 16.09.2025: self-certified CTE up to ₹50 L; CTO up to the applicable ₹25–50 L slab) and within the **micro-enterprise** band. Land is the promoter's own and is excluded (in-kind), so it does not consume the ₹50 L headroom.

7.3 Consent path

- **Green + $\leq ₹50$ L → self-certification-based CTE and CTO** — fast, low-fee, no public hearing.
- **No Environmental Clearance** (biochar from agri-biomass is not an EIA-2006 scheduled activity; not Red).
- A **written KSPCB clarification** confirming the Green entry for this unit has been sought (see the separate KSPCB category-clarification letter); sanction should follow that confirmation.

7.4 Honest caveats (to close before sanction)

- **Clubbing risk.** Pollution boards may **aggregate co-located, commonly-owned units** and assess them as one. To keep each unit independently $\leq ₹50$ L and Green, units should be **genuinely separate** — distinct premises/demarcated blocks, separate Udyam registrations, separate metering and consent applications — and this treatment should be **confirmed in writing with KSPCB**. The Green classification itself does **not** depend on the split; the split only unlocks self-certification and micro-benefits.
- **Category base.** Confirm whether KSPCB's ₹50 L test is on *plant & machinery* or *total fixed capital*, and size accordingly.

8. Machinery & suppliers (per micro-unit)

Local (Coimbatore-Pollachi hub, 30-60 km — preferred for service/spares): coconut-shell/biomass carbonization plant manufacturers (e.g., Micro Fab Engineers, Essar Engineers, Veera Biopower and similar). Indicative small semi-auto units 1 t/day-class ≈ ₹6-25 L.

National (continuous, low-emission, certification-grade — better for EBC/carbon): - **Kerone Engineering (Mumbai)** — continuous multi-feedstock pyrolysis, waste-heat drying, cyclone + scrubber, SCADA. - **W2E Innovations (Ahmedabad/Kharagpur; Bengaluru branch)** — integrated biomass→biochar + wood-vinegar + fertilizer, "verification-ready." - **Susstains Engineering (Karnataka)** — IIT-Madras-derived zero-smoke continuous process (~34% yield). - **Ylem Energy (New Delhi)** — smaller batch furnaces (lower cost; batch → less consistent).

Recommendation: for a micro-unit, buy a **compact continuous/semi-continuous ~0.5 t/day plant with integrated dryer + afterburner + scrubber** (not a bare batch kiln) so stack emissions comfortably meet the Green/guideline norms. Obtain quotes from one local Coimbatore/Pollachi maker (price + service) and one national continuous-tech supplier; compare on **emission compliance and yield**. Budget ~₹22 L for the reactor + gas train + dryer per unit (see Annexure D).

9. Land, layout & civil works

Land: promoter-owned **38-cent plot** (~1,538 sq m) + adjacent farm (feedstock/drying). The plot comfortably hosts up to **three compact micro-units** with shared greenbelt, road and gate; each unit is a **separately demarcated block** (supporting independent consent).

Indicative per-unit footprint (compact):

Zone (per unit)	Area (sq m)
Pyrolysis + emission control	60
Feedstock day-bin (1-2 day)	30
Enrichment / curing	90
Product storage + bagging	60
Share of greenbelt / roads / utilities	~120
Per-unit total	~360

Pre-checks: confirm plot land-use is convertible to industrial where required; KSPCB setback from residences/water bodies/eco-sensitive zones; road access; 3-phase power; water availability. **Construction per unit (compact):** process shed + curing/storage + site works ≈ **₹12 L** (see cost table).

10. Regulatory & statutory compliance (per unit)

Approval	Authority	Notes
Udyam (MSME) registration	Online	Separate registration per unit; gateway for schemes
Consent to Establish (CTE)	KSPCB	Green category, self-certification route (≤ ₹50 L)
Consent to Operate (CTO)	KSPCB	Green; self-certification per applicable slab
Panchayat D&O (Dangerous & Offensive) licence	Grama Panchayat	Manufacturing unit
Building permit	Grama Panchayat	Shed/plant
Fire NOC	Fire & Rescue Dept.	Thermal process
GST registration		
Land-use conversion	Revenue Dept.	If plot is agricultural

Operating without CTE/CTO is a criminal offence under EPA 1986. No open burning of biomass at any stage. Environmental Clearance is not required (non-EIA-scheduled, non-Red).

11. Cost of project (per micro-unit)

#	Item	₹ Lakh
1	Site development + compact process shed + curing/storage (civil)	12
2	Pyrolysis reactor ~0.5 t/day (continuous/semi-continuous, multi-feedstock)	16
3	Emission control — afterburner/flare + cyclone + scrubber + stack (Green-compliant)	6
4	Char cooling, sieving, enrichment mixer, bagging	5

#	Item	₹ Lakh
5	Electrical, water, fire, weighing	4
6	Feedstock drying floor + chipper/shredder	3
7	Lab / QC + office corner	1
8	Preliminary & pre-operative (self-cert CTE/CTO, DPR, EBC/MRV)	1
	Fixed capital sub-total (per unit)	48
9	Margin money for working capital	4
	Total project cost (per unit)	~₹52 lakh

Land is the promoter's own (in-kind) and is **excluded** from the fixed-capital figure that is tested against the ₹50 L self-certification threshold. **Aggregate for 3 units ≈ ₹1.44 cr fixed capital** (units funded sequentially).

12. Means of finance & subsidy stack (per unit)

Source	₹ Lakh	%
Promoter equity (cash)	14	27%
Term loan — bank, under AIF (3% interest subvention, 7 yrs), CGTMSE collateral-free	34	65%
Working capital limit (bank cash credit)	4	8%
Total	52	100%

Kerala ESS capital subsidy (~₹20 L per unit) — at the ~45% band on ₹48 L eligible fixed capital, within the ₹40 L cap — is received as **reimbursement after commissioning** and applied to **prepay the term loan**, cutting effective per-unit debt to ~₹14 L.

Subsidy stack logic

- **Anchor = Kerala ESS** (state capital subsidy). Target **45%** band: 25% base (young/women/SC-ST/NRK) + 10% priority sector (agro-based / bio-fertilizer) + 10% new technology. *Palakkad does not get the extra 10% district bonus.*
- **+ AIF** (central): 3% interest subvention + CGTMSE guarantee — different basis, **stacks legitimately with ESS**; confirm convergence in writing with DIC + bank.
- **+ CGTMSE**: collateral-free loan → the farm is not mortgaged.
- **Multi-unit ESS**: confirm with DIC whether each independently-registered micro-unit is **separately eligible** for ESS; if units are clubbed, the aggregate stays within scheme caps.
- **Do NOT use PMEGP** — its rules disqualify units that have availed other government subsidy, voiding ESS + AIF.
- **Effective term-loan interest**: 9% – 3% = **~6%**.

13. Working capital (indicative, annual, per unit)

Feedstock collection/purchase/transport, enrichment inputs (poultry manure, neem cake, rock phosphate, microbial consortium), packaging, wages, power, and ~1-2 months finished-goods holding. Per-unit operating cost ≈ **₹30 L/yr.**

14. Profitability & ROI (per unit and aggregate)

Common assumptions: 300 operating days; biochar 1 t ≈ 2.2 t CO₂; carbon net realization ~₹10,000/t CO₂ (after aggregator/MRV); enriched blended price ₹35/kg. Effective interest 6%.

Scenario	Biochar	Revenue	Op. cost	Net profit (approx)	Payback
Per unit — base (physical only)	150 t	₹52.5 L	₹30 L	~₹12 L	~2.5-3.5 yr
Per unit — + carbon	150 t	₹52.5 L + ₹33 L = ₹85.5 L	₹32 L	~₹40 L	~1.2-1.5 yr
3 units — base (physical only)	450 t	₹1.58 cr	₹0.90 cr	~₹36 L	—
3 units — + carbon	450 t	₹1.58 cr + ₹0.99 cr = ₹2.57 cr	₹0.96 cr	~₹1.2 cr	<1.5 yr

Net capital at risk (per unit) ≈ ₹28 L after ESS ~₹20 L. **ROI (per unit, base): ~20%; with carbon: ~85%.** Carbon revenue is upside — treat physical-only as the bankable base case; carbon requires certification + a signed aggregator offtake.

Drivers: low-cost multi-source feedstock, ~6% subsidised interest, ESS, selling enriched (not raw) product, and staged low-risk capital.

15. Carbon-credit strategy & aggregator options

At ~330 t CO₂/yr per unit (~990 t CO₂/yr at 3 units), volumes are **too small to transact directly with corporate buyers** (~85% of CORCs are sold via brokers/aggregators). Route through an **aggregator/developer** who provides MRV, certification (Puro.earth CORC methodology) and buyer access. Aggregating the 3 units' output improves the transactable volume.

Party	Role / fit
Varaha (Gurugram/New Delhi) — India's largest biochar CDR developer; Puro.earth + Verra; offtakes with Google, Microsoft, Kellanova	Best first contact. Its Industrial Partners Program lets facilities with feedstock + conversion capability use Varaha's digital MRV + credit commercialization and issue CORCs.
Equilibrium — industrial biochar operator	Developer/partner; potential offtake channel.
Eco Saarthi — cooperative aggregating agri residue → biochar	Fit for an aggregation structure.
Puro.earth	The registry/standard (CORCCHAR methodology); certification via a developer.
BioFlux (advisory)	Independent biochar-CDR advisory — feasibility, MRV & certification, buyer alignment.
Marketplaces/brokers — Supercritical, Senken	Secondary sale/price-discovery.

Action: approach **Varaha first** for an MRV + offtake partnership; target a **Letter of Intent (LOI)** before scaling to bank the carbon revenue. Document feedstock as **residual/waste biomass** for additionality.

16. Value addition — enriched bio-fertilizer

Base biochar is K-rich; enrich to a balanced slow-release organic input: - **Nitrogen/phosphorus:** poultry manure, neem cake, bone meal / rock phosphate. - **Biology (branded premium):** microbial consortium — Azospirillum/Azotobacter (N-fix), PSB/phosphobacteria, potash mobilizers, Trichoderma, mycorrhiza (VAM). - **Base/biology co-compost:** vermicompost + cow dung; Kerala liquid organics (fish amino acid, panchagavya, jeevamrutham); seaweed extract for micronutrients. - **Methods:** nutrient impregnation/soaking then low-temp drying; or co-composting/vermicomposting with ~10% biochar; pelletize to cut leaching. - **Certifications for premium/export:** EBC, IBI, organic-input (Jaivik Bharat / NPOP for export).

17. Implementation schedule (sequence, not calendar)

1. Feedstock lab proximate analysis (moisture, fixed carbon, ash, K) — decides final sizing.
2. Udyam registration (Unit-1); DIC/ESS pre-consultation; feedstock MoUs with 3–4 local sources.
3. **Written KSPCB categorisation confirmation** (Green) for the unit.
4. **Self-certified CTE (Green);** machinery quotes & order (~0.5 t/day reactor + gas train + dryer).
5. Compact civil construction + plant erection.
6. Panchayat D&O + building permit + fire NOC; **self-certified CTO.**
7. AIF term loan + CGTMSE; commissioning; ESS claim on installed assets.
8. Product validation on own farm; EBC/organic certification.
9. Engage carbon aggregator (Varaha); secure LOI.
10. **Replicate Unit-2 and Unit-3** from cash flows / fresh micro-loans (repeat steps 2–7 per unit).

18. Risks & mitigation

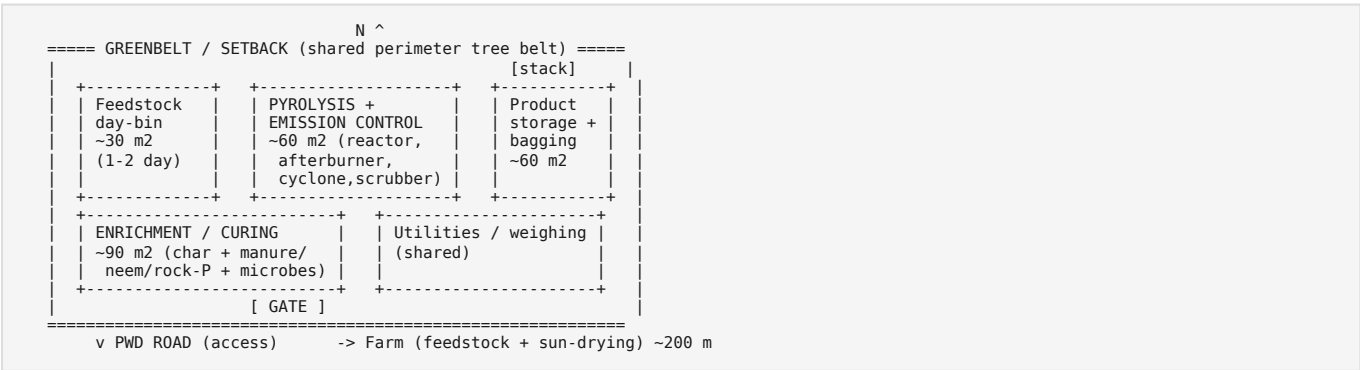
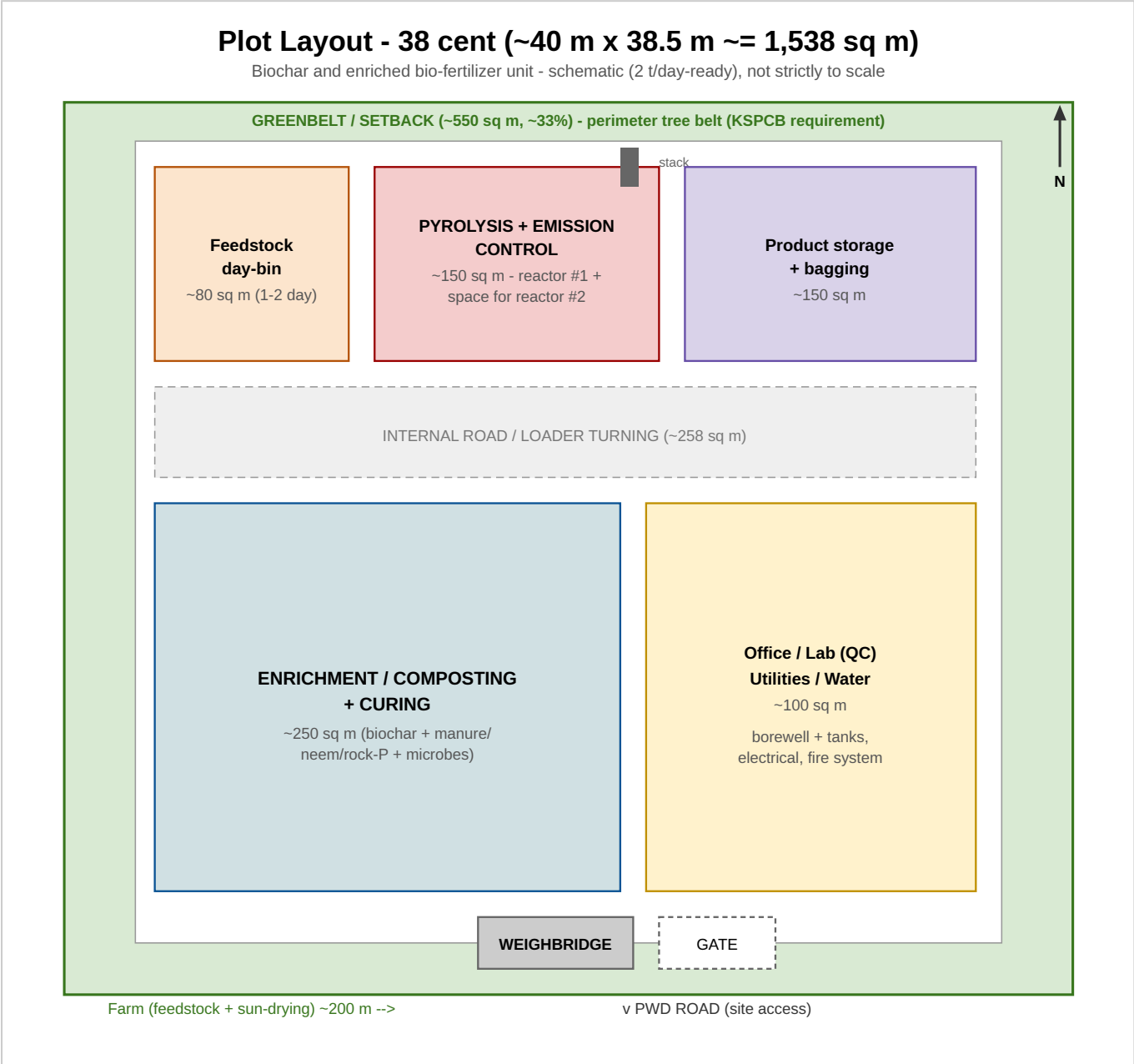
Risk	Mitigation
Category/clubbing — KSPCB may club co-located commonly-owned units	Keep units genuinely independent (separate blocks, Udyam, metering, consents); secure written Green confirmation ; Green rests on the process, not the split
Feedstock moisture / drying energy	Sun-dry at farm; waste-heat dryer; lab-test first
Feedstock security	Multi-source (own farm + 3–4 open-market suppliers) — no single-supplier dependency
Demand-side weakness (Indian biochar)	Own-farm consumption; sell enriched/branded; export the credit
Carbon-credit dependency	Bank case = physical only; carbon = upside pending LOI
Multi-unit subsidy eligibility	Confirm per-unit ESS eligibility with DIC in writing
Product consistency for certification	Continuous reactor + fixed blend recipe

19. Assumptions & disclaimer

All costs, yields, prices and returns are **planning-grade indicative estimates** (mid-2026 market references) for scheme application and bank appraisal. They must be confirmed by: (a) a laboratory proximate analysis of the actual feedstock; (b) formal machinery quotations; (c) written confirmation of ESS/AIF eligibility and convergence from DIC Palakkad and the lending bank; (d) a **written KSPCB categorisation (Green) and self-certification confirmation**, including treatment of multiple co-located units; and (e) a carbon-credit LOI from an aggregator before carbon revenue is counted. Regulatory categories and subsidy rates are subject to the prevailing KSPCB, DIC and scheme guidelines at the time of application.

Annexure A — Micro-unit layout & site plan

plot-layout.svg (open in any browser) shows the owned site accommodating the phased units. Text schematic of a **single ≤ ₹50 L micro-unit** below:



Notes: each unit is a **separately demarcated block** with its own reactor, emission-control train and storage (supporting independent Green/self-certified consent); bulk biomass is stored **off-site at the farm** (200 m), so only a 1-2 day day-bin sits in the unit. Up to three such blocks share the greenbelt, gate and utilities on the 38-cent plot.

Annexure B — Term-loan amortisation (per unit)

Assumptions: term loan ₹34 L per unit; effective rate **6% p.a.** (9% – 3% AIF subvention); tenure **7 years**; **6-month construction moratorium** (interest served, no principal in Year 1); **Kerala ESS ~₹20 L received ~Month 12 and applied as principal prepayment**, leaving ~₹14 L to amortise over Years 2–7 (EAI ≈ ₹2.85 L/yr, ≈ ₹0.24 L/month).

Year	Opening (₹L)	Interest @6% (₹L)	Principal repaid (₹L)	ESS prepayment (₹L)	Closing (₹L)
1	34.00	2.04	0.00 (moratorium)	20.00	14.00
2	14.00	0.84	2.01	–	11.99
3	11.99	0.72	2.13	–	9.86
4	9.86	0.59	2.26	–	7.60
5	7.60	0.46	2.39	–	5.21
6	5.21	0.31	2.54	–	2.67
7	2.67	0.16	2.69	–	~0.00
Total		~5.1	~14.0	20.0	

AIF benefit: the 3% subvention saves ~₹2.5 L of interest per unit over the loan life versus a 9% loan; the loan is **collateral-free** under CGTMSE — the farm is not mortgaged.

Annexure C — Month-by-month cash flow, Year 1 (per unit ramp)

Assumptions (physical-sales base case): commissioning at **Month 5**; ramp 30% → 100% by Month 10. Full-capacity monthly revenue **₹4.38 L** (₹52.5 L/yr); operating cost ≈ **₹1.2 L fixed + ₹1.3 L variable at 100%**; interest served during moratorium ≈ **₹0.17 L/month**. Capex is funded by equity + term loan (not shown in operating flow). ESS ₹20 L inflow (~Month 12) is a financing item applied to loan prepayment (Annexure B).

Month	Phase	Capacity	Revenue (₹L)	Op. cost (₹L)	EBITDA (₹L)	Interest (₹L)	Net op. cash (₹L)
1–4	Construction / pre-op	–	0.00	trial only	–	0.17/mo	interest served from loan
5	Commissioning	30%	1.31	1.59	–0.28	0.17	–0.45
6	Ramp	45%	1.97	1.79	0.18	0.17	0.01
7	Ramp	60%	2.63	1.98	0.65	0.17	0.48
8	Ramp	75%	3.28	2.18	1.10	0.17	0.93
9	Ramp	90%	3.94	2.38	1.56	0.17	1.39
10–12	Full	100%	4.38/mo	2.50/mo	1.88/mo	0.17	1.71/mo

Steady state (Year 2+, per unit): monthly revenue ₹4.38 L – opex ₹2.50 L = **EBITDA ₹1.88 L**; less EMI ₹0.24 L → **net cash ~₹1.64 L/month (~₹19.6 L/yr)** before tax and depreciation.

Debt-service coverage (Year 2, per unit): EBITDA ≈ ₹22.5 L ÷ debt service ≈ ₹2.85 L → **DSCR ≈ 7.9** (very comfortably bankable). Even before ESS prepayment, DSCR on the full ₹34 L loan is well above 2.0.

Upside: adding realized carbon revenue (~₹33 L/yr per unit) lifts EBITDA sharply and shortens payback to well under ~1.5 years — kept out of the base case above pending a signed aggregator LOI.

Annexure D — Machinery quotation-comparison template (fillable, per micro-unit)

How to use: send this to at least **two local vendors (Coimbatore/Pollachi hub)** and **one national continuous-technology vendor**. Ask each to quote a complete **~0.5 t/day micro-unit** (reactor + dryer + emission train + downstream), state guaranteed emission values and yield, then carry the total into **Section 11 — Cost of project**.

Table D-1 — Scope & required specification (issue to vendors)

#	Equipment	Sized for	Required specification / notes	→ Cost item
1	Pyrolysis / carbonization reactor	~0.5 t/day (~100 kg/hr feed)	Continuous or semi-continuous; multi-feedstock (husk, frond, shell, coir, sawdust, shell); yield ≥25–30% dry basis	#2
2	Feed system (hopper, screw/belt, metering)	0.5 t/day	Variable-speed feed; suited to shredded mixed biomass	#2

#	Equipment	Sized for	Required specification / notes	→ Cost item
3	Waste-heat feed dryer	0.5 t/day	Uses pyrolysis/syngas waste heat; inlet moisture up to ~25%	#2
4	Emission control — afterburner/flare + cyclone + alkaline scrubber + mist eliminator + ID fan + stack	0.5 t/day	Must meet KSPCB Green / 02.12.2025 guideline stack norms; vendor to state guaranteed emission values & stack height	#3
5	Char cooling / quench + discharge	0.5 t/day	Sealed cooling; moisture/temperature controlled	#4
6	Sieving / screening	0.5 t/day	Grade separation (powder/granular)	#4
7	Enrichment mixer / blender	0.5 t/day	For char + manure/neem/rock-P + microbial slurry	#4
8	Weighing + bagging	0.5 t/day	25 kg bags + FIBC jumbo option	#4
9	Control panel / instrumentation	0.5 t/day	Temp/pressure monitoring, interlocks, e-stop	#5
10	Erection, commissioning & operator training	—	Include site supervision & trial run	#2–#5
11	Freight, insurance, unloading	—	To site, Eruthenpathy	#2–#5
12	Taxes (GST)	—	State applicable rate separately	—

Table D-2 — Price comparison (fill from vendor quotes, ₹)

#	Equipment (as above)	Vendor A: _____	Vendor B: _____	Vendor C: _____
1	Pyrolysis reactor (0.5 t/day)	₹_____	₹_____	₹_____
2	Feed system	₹_____	₹_____	₹_____
3	Waste-heat dryer	₹_____	₹_____	₹_____
4	Emission control train	₹_____	₹_____	₹_____
5	Char cooling / quench	₹_____	₹_____	₹_____
6	Sieving / screening	₹_____	₹_____	₹_____
7	Enrichment mixer	₹_____	₹_____	₹_____
8	Weighing + bagging	₹_____	₹_____	₹_____
9	Control panel	₹_____	₹_____	₹_____
10	Erection, commissioning, training	₹_____	₹_____	₹_____
11	Freight, insurance	₹_____	₹_____	₹_____
	Sub-total (ex-GST)	₹_____	₹_____	₹_____
12	GST	₹_____	₹_____	₹_____
	TOTAL (incl. GST)	₹_____	₹_____	₹_____
	<i>Target ≈ ₹22 L for reactor+dryer+emission (items #2–#3 of Section 11)</i>			

Table D-3 — Commercial-terms comparison (fill from vendor quotes)

Parameter	Vendor A	Vendor B	Vendor C
Make / model offered			
Rated capacity (kg/hr feed; t/day biochar)			
Guaranteed biochar yield (% dry basis)			
Material of construction (reactor)			
Connected electrical load (kW)			
Guaranteed stack emissions vs KSPCB Green/guideline norms			
Feedstock types accepted (husk/frond/shell/coir/sawdust)			
Delivery period (weeks from PO)			
Warranty (months)			
After-sales / spares — distance from site			
Payment terms			
Training & commissioning included?			
References / plants in operation (contactable)			

Evaluation guidance: do not select on price alone — weight **emission-compliance (for Green CTO), guaranteed yield, feedstock flexibility, and after-sales proximity** (favouring the Coimbatore/Pollachi hub for fast service). A clean, low-emission continuous micro-unit both satisfies the Green/guideline norms and produces consistent, certifiable biochar.